

CORINNA (YUXIN) MA

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Research Interests

Algorithmic Game Theory • Combinatorial Optimization • Algorithm Design • Computational Social Choice

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Industrial & System Engineering Department

August 2025-Expected, May 2027

- **Selected Coursework:** Foundations of Optimization, Network Flows & Combinatorial Optimization, Large-Scale Optimization & Machine Learning, Predictive Analytics; Planned: Approximation Algorithms, Stochastic Optimization

University of Toronto

Toronto, ON

Honours Bachelor of Science with Distinction

September 2021-June 2025

- **Specialist:** Mathematics & Its Applications · **Major:** Physics
- **Selected Coursework:** Analysis & Real/Complex Analysis, Groups & Symmetry, Algebra, Probability & Statistics, Theory of Computation, Data Structure & Analysis, Geometry of Curves & Surfaces, Ordinary & Partial Differential Equations, Time Series Analysis, Classical Mechanics, Quantum Physics, Electromagnetic Theory, Computational Physics, General Relativity
- **Awards:** Dean's List Scholar

RESEARCH EXPERIENCE

University of Southern California

Los Angeles, CA | 2025 - Present

Research Assistant, Computational Social Choice & Algorithmic Game Theory

Advisor: Prof. David Kempe

- Studied metric distortion under misinformed preferences, developing an α -misinformation framework that relaxes the assumption that voters rank candidates exactly according to their underlying metric distances
- Analyzed how misinformation affects distortion guarantees of voting rules using combinatorial constructions and structural arguments; current work includes theoretical analysis of different voting rules
- Investigated algorithmic questions in delegation and liquid democracy, using graph-based optimization, linear programming duality, and network-flow techniques to establish theoretical guarantees
- Manuscript: Corinna Ma, Grayson York, David Kempe. *Democracy in Darkness: Metric Voting Under Worst-Case Misinformation* – In submission

Research Assistant, Optimization & Numerical Methods

Advisor: Prof. T.K. Satish Kumar

- Developed optimization-based methods for dynamical systems using Bézier-curve parameterizations and quadratic programming, replacing repeated simulation with direct optimization over system trajectories
- Developed Laplace-transform-based formulations for incorporating ODE dynamics into optimization models
- This work resulted in *Bypassing Simulation-Based Methods: A Case Study in HVAC Scheduling*, published at ISAIM 2026

Research Assistant, Algorithms & Combinatorial Optimization

Advisor: Prof. John Gunnar Carlsson

- Studied approximation algorithms for clustering and NP-hard graph optimization problems, emphasis on performance guarantees
- Investigated algorithmic problems in stable matching and discrete optimization, including structural properties that enable efficient or approximate solution

Earlier Research

Cathaypath Institute of Science

Remote | 2020

Researcher, Constructive Analysis

Research mentorship with Prof. Vladimir Chernov, Dartmouth College

- Studied nice open covers in constructive analysis, investigating conditions under which restricted classes of open covers exhibit compactness-like properties
- Developed mathematical arguments concerning the existence and properties of nice open covers within a constructive framework
- Co-authored *A Study on Nice Open Covers in Constructive Analysis* (Cheng, Gao, Ma, Ning & Xu, 2020), arXiv:2009.02727.

Yau Tsinghua Math Camp

Beijing, China | 2020

Research Participant, Algebraic Combinatorics

- Undertook intensive study in linear algebra, algebraic combinatorics, analysis and topology, with an emphasis on proof-based problem solving and exposure to modern mathematical research
- Conducted mentored research on algebraic combinatorics and generator sets, and developed discrete mathematical constructions

INDUSTRY EXPERIENCE

Eikon Therapeutics

May 2026-Present | San Francisco, CA

Software Engineer & Computational Modeling Intern

- Developed optimization methods for image-tracking pipelines using image fingerprinting and the Lin–Kernighan heuristic to improve tracking accuracy and computational performance
- Developed probabilistic models and inference algorithms for biological image analysis using Bayesian inference methods in JAX and NumPyro, with emphasis on scalable statistical learning and computational efficiency
- Design, implement, and benchmarked Bayesian inference methods—including variational inference, variational EM, Markov chain Monte Carlo (MCMC), and Gibbs sampling—for scalable parameter estimation and uncertainty quantification

Dalian Commodity Exchange

May 2024-July 2024 | Dalian, China

Data Analyst Intern

- Analyzed 10+ years of multi-asset futures data to characterize volatility, cross-asset dependence, and price dynamics using statistical and time-series methods
- Developed regression and time-series models to identify factors associated with major commodity price movements and evaluate their explanatory power across market regimes
- Built reproducible pipelines for large-scale data processing and statistical analysis, enabling systematic evaluation of empirical hypotheses across commodity markets

LEADERSHIP & TECHNICAL ACTIVITIES

QuantSC

Sep 2025-Aug 2026| Los Angeles, CA

Quantitative Researcher → Project Lead

- Developed machine learning pipelines for financial time-series forecasting using on-chain and market data; engineer predictive features, evaluate systematic strategies, and analyze model performance through backtesting
- (Project Lead since Jan 2026) Designed and implemented pairs trading strategies using LSTM models to capture temporal dependencies and mean-reversion signals; applied k-means clustering to identify correlated asset groups and detect lead–lag relationships for statistical arbitrage opportunities

SELECTED TALKS

Democracy in Darkness: Metric Voting Under Worst-Case Misinformation

Sep 2026

USC Theory Lunch, University of Southern California

- Presented a relaxed metric-distortion framework for voting under α -misinformed preferences, together with a flow-based approach for analyzing the robustness of voting rules
- Presented distortion bounds for Plurality, Borda, Plurality Veto, Copeland, STV, and single-elimination tournaments, illustrating how robustness under misinformation can differ substantially from guarantees in the classical metric model

Optimization Strategies for Airplane Boarding

Apr 2025

University of Toronto

- Modeled airplane boarding as a geometric optimization problem, using a Lorentzian-geometry framework to characterize passenger interactions and boarding dynamics
- Analyzed alternative boarding strategies through the geometric model to investigate how passenger ordering affects overall boarding efficiency
- Presented the project and its mathematical formulation in a course presentation

TECHNICAL SKILLS

- Programming: Python, C++, Lean, Go, Java, TypeScript, Solidity
- Optimization/Scientific Computing: Gurobi, NetworkX, JAX, NumPy/SciPy, NumPyro
- Machine Learning & Statistical Methods: Bayesian Inference, Variational Inference, MCMC, Gibbs, Numerical Optimization, LP